



Dr. Romain Mottier

Computational Scientist — PhD in Applied Mathematics

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French (native) • English (C2) • Valid Swiss work permit (G)

PROFILE

Computational Scientist in Applied Mathematics Physics. Five years developing and shipping simulation code for engineering and physics problems: wave propagation, solid–fluid interaction, geophysics, electromagnetics, planetary thermal modelling. **C++**, **Fortran** and **Python** at production level, from **PDE solver design** and mathematical validation through to large-scale parallel runs (**MPI/OpenMP**) on HPC clusters. Contributor to a production geophysics code used for seismic simulation.

Position sought. R&D and scientific engineering roles where rigour translates into industrial applications. Comfortable entering a new physical domain from scratch: four distinct fields in five years, each carried through to working, validated code.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher — Computational Mathematics 10/2025 – present
University of Basel, Dept. of Mathematics and Computer Science Basel, Switzerland

- Extending **DiSk++** (C++17 finite element library) with explicit multilevel local time-stepping for high-order wave propagation solvers.
- Developing high-order solvers for non-linear PDEs with variable Sobolev exponents.
- Two research articles in preparation & three talks at international conferences in 2026 (Delft, Vienna, Montreal).

PhD Researcher — Applied Mathematics & Scientific Computing 10/2021 – 07/2025
French Atomic Energy Commission (CEA) & École nationale des Ponts et Chaussées (ENPC) Paris, France
& Polytechnic Institute of Paris (IP Paris)

- Developed and analysed **new high-order finite element methods** for coupled **elasto-acoustic wave propagation** — from mathematical analysis through to production code.
- Ran large-scale 2D/3D simulations on HPC cluster for geophysical wave-propagation.
- Authored **3 research articles** published in leading international journals and presented at **5 international conferences** in Europe and North America.
- Collaborated across three institutions (universities and CEA), reporting to both academic and industrial supervisors.

Research Intern — Computational Electromagnetics 03/2021 – 08/2021
ONERA, The French Aerospace Lab Toulouse, France

- Implemented Compatible Discrete Operator (CDO) schemes for the time-domain Maxwell equations, benchmarked against reference solutions.

Research Intern — Planetary Thermal Modelling 05/2020 – 08/2020
European Space Agency (ESA/ESTEC) Noordwijk, Netherlands

- Implemented and solved a numerical heat-transfer model computing the surface and sub-surface temperature distribution on Mercury.

SOFTWARE PROJECTS

DiSk++ github.com/wareHHouse/diskpp

C++17 template library for discontinuous skeletal methods (HHO/HDG)

- Contributed the **elasto-acoustic wave propagation module** to an open-source scientific library. Implemented templated C++ covering assembly, static condensation and explicit/implicit time integration, dimension- and mesh-agnostic by design — supporting the results published in **ESAIM: Mathematical Modelling and Numerical Analysis** (2025) and **Computer Methods in Applied Mechanics and Engineering** (2026).
- Built on Eigen, MUMPS, SILO/HDF5, gmsh and CMake; validated against analytical solutions on 2D/3D benchmarks up to order 7, with problems of several million degrees of freedom.

SEM3D github.com/sem3d/SEM

Fortran 90 spectral-element solver for 3D seismic wave propagation

- Contributed to a **production geophysics code** jointly maintained by CEA, IPGP, CentraleSupélec and CNRS, used for broadband (0–10 Hz) regional-scale earthquake simulation at nuclear sites. Implemented the HDG method for the elastodynamic wave equation, with explicit time integration and MPI/OpenMP parallelisation, and ran large-scale simulations on cluster EXA1.

ProtoN github.com/wareHHouse/ProtoN

C++ library for prototyping cut/unfitted finite element methods

- Implemented discontinuous cut-FEM (unfitted HHO) schemes with polynomial-extension stabilisation for elliptic interface problems, supporting the results published in **SIAM Journal on Numerical Analysis** (2026).

TECHNICAL SKILLS

Programming	C++ • Python • Fortran • MATLAB • Bash
Applied mathematics	Partial Differential Equations • Numerical Analysis • Linear Algebra • Optimisation
Scientific computing	High-Performance Computing (MPI, OpenMP) • Eigen • Intel MKL • BLAS/LAPACK
Numerical methods	Finite Element & Finite Volume Methods • Time Integration • Sparse Linear Solvers
Data & analysis	NumPy • Pandas • Matplotlib • Monte-Carlo • ParaView
Software engineering	Git • CMake • Unit testing • Code review • Linux • LLM coding assistants (Claude Code, Copilot)
Application domains	Wave Propagation & Solid-Fluid Interaction • Geophysics • Electromagnetics

EDUCATION

PhD in Applied Mathematics & Scientific Computing	10/2021 – 07/2025
French Atomic Energy Commission (CEA) & École nationale des Ponts et Chaussées (ENPC)	Paris, France
<i>Polytechnic Institute of Paris (IP Paris)</i>	
MSc in Engineering — Modelling and Fluid–Structure Computation	09/2018 – 09/2021
Université de Toulon, SeaTech Engineering School	Toulon, France
• Exchange year — <i>MSc Numerical Methods in Engineering</i> , Universitat Politècnica de Catalunya (UPC), Barcelona, Spain (09/2020 – 02/2021).	

TEACHING

Lecturer — Numerical Analysis of ODEs	Fall 2026
University of Basel — final-year BSc course, Dept. of Mathematics and Computer Science	Basel, Switzerland
Supervision of Master’s semester projects	Fall 2025 – Spring 2026
University of Basel — <i>Numerical Methods for PDEs</i>	Basel, Switzerland
• Supervision of Master’s students on finite element methods for linear elasticity (fall semester) and on the spectral element method (SEM) for the acoustic wave equation, adapted to industrial HPC architectures .	
Teaching Assistant — Numerical Methods	Spring 2023
Paris Dauphine University — BSc Mathematics & Computer Science	Paris, France
• Lectures and lab sessions taught in Python (NumPy, Jupyter): nonlinear equations, polynomial interpolation, quadrature, direct and iterative linear solvers, eigenvalue computation.	
Teaching Assistant — Numerical Methods	Fall 2022
Sorbonne University — MSc Computational Mechanics	Paris, France
• Lab sessions and numerical modelling projects: linear systems, finite differences, continuum mechanics.	

PUBLICATIONS, PREPRINTS & CONFERENCES

- Unfitted hybrid high-order methods stabilized by polynomial extension for elliptic interface problems. *with E. Burman and A. Ern* — published in **SIAM Journal on Numerical Analysis**, 2026 [↗](#)
- Hybrid high-order methods for elasto-acoustic wave propagation in the time domain. *with A. Ern, R. Khot and L. Guillot* — published in **ESAIM: Mathematical Modelling and Numerical Analysis**, 2025 [↗](#)
- Elasto-acoustic wave propagation in geophysical media using hybrid high-order methods on general meshes. *with A. Ern and L. Guillot* — published in **Computer Methods in Applied Mechanics and Engineering**, 2026 [↗](#)
- Multilevel explicit local time-stepping for wave propagation with high-order hybrid nonconforming schemes (HDG/HHO). *with M. J. Grote* — *in preparation*
- A hybrid high-order method for variable-exponent Leray–Lions problems. *with A. Harnist* — *in preparation*

International & European Conference presentations

Talks — Wave-NED (Delft, 08/2026) • CMAM (Vienna, 07/2026) • WAVES (Montreal, 06/2026) • WCCM (Vancouver, 07/2024) • ECCOMAS (Lisbon, 06/2024) • CANUM (Île de Ré, 05/2024).

Posters — AGU Fall Meeting (San Francisco, 12/2023) • CJCMA (Paris, 09/2023).

REFERENCES

Prof. Marcus J. Grote — Postdoctoral advisor Professor, University of Basel marcus.grote@unibas.ch	Dr. Laurent Guillot — Industrial PhD advisor Researcher, CEA laurent.guillot.blr@gmail.com
Prof. Alexandre Ern — PhD advisor Senior Researcher, INRIA (SERENA) alexandre.ern@enpc.fr	Dr. Guillaume Delay Researcher, Sorbonne University (LJLL) guillaume.delay@sorbonne-universite.fr